

Beyond Firewall: Leveraging Machine Learning for Real-Time Insider Threat Identification and User Profiling

INTRODUCTION TO INSIDER THREAT

Insider threat means:

A threat caused by authorized users such as:

- Employees
- Contractors
- Administrators

Types:

- Malicious insider -A person inside an organization who intentionally causes harm or steals information.
- Negligent insider -A person inside an organization who causes security problems by mistake or carelessness
- Compromised insider -A person inside an organization whose account or system is hacked or controlled by an attacker.

LIMITATIONS OF TRADITIONAL SECURITY

Traditional tools:

- Firewalls
- Antivirus
- Signature-based systems

Problems:

- Detect mainly external attacks
- Cannot analyze behavior deeply
- Slow detection
- High false alarms

*Traditional cybersecurity tools are mainly designed to stop threats coming from outside the organization.

*For example, firewalls block unauthorized external access, and antivirus detects known malware signatures.

OBJECTIVE OF THE PAPER

The paper aims to:

- Detect insider threats instantly
- Monitor user behaviour continuously
- Profile users dynamically
- Assign risk levels

Risk levels:

- Low
- Medium
- High

*The main objective of this paper is to build an intelligent system that continuously observes employee activities and predicts risk levels in real time.

PROPOSED METHODOLOGY

Add a little bit Data Collection

- Feature Extraction
- User Profiling
- ML Classification
- Risk Prediction
- Alert Generation

Models used:

- Logistic Regression -A machine learning method used to predict yes/no or true/false outcomes.
- Decision Tree -A method that makes decisions step by step like a flowchart to classify or predict data.
- Random Forest -A method that uses many decision trees together to make more accurate predictions.
- XGBoost -An advanced machine learning method that improves prediction accuracy by learning from previous mistakes.

RESULTS

Best Model: XGBoost

Why best:

- High accuracy
- Better precision
- Faster prediction
- Strong classification ability

Metrics: Metrics are measurements used to check how well a model works.

- Accuracy -How many answers are correct.
- Precision -How many predicted results are right.
- Recall -How many actual positive cases the model correctly finds.
- F1-score -A score that checks both precision and recall together.

RELATION TO SURVEY PAPER

Survey challenge → Solution in this paper

Weak real-time detection

→ Real-time ML monitoring

Static detection

→ Dynamic profiling

Poor behavior analysis

→ User profiling

Traditional systems

→ Intelligent classification

LIMITATIONS

Limitations are:

- Synthetic dataset -Artificially created data used for testing instead of real-world data.
- Limited real-world testing -A system is not tested enough in actual real-life situations
- No psychological factors -Human emotions, behavior, or mental reasons are not considered in the study or system.
- Limited explainability -The system gives results, but it is hard to understand how or why it made that decision.

FUTURE WORK

Possible improvements:

- Real CERT dataset
- Human behaviour indicators
- Explainable AI
- Large-scale enterprise validation

CONCLUSION

Paper proves:

- ML improves detection
- Real-time profiling works
- XGBoost performs best
- Behavioral AI is future security direction